CR Geometry

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Chapter 1 Symplectic Manifolds

§ 1.1 Symplectic Manifolds

Definition 1.1.1. By a **symplectic manifold** we mean an even-dimensional differentiable (C^∞) manifold M^{2n} together with a global 2-form Ω which is closed and of maximal rank, i.e., $d\Omega = 0, \Omega^n \neq 0$.

By a **symplectomorphism** $f : \left(M_1, \Omega_1\right) \longrightarrow \left(M_2, \Omega_2\right)$ we mean a diffeomorphism $f : M_1 \longrightarrow M_2$ such that

$$f^*\Omega_2 = \Omega_1.$$

Definition 1.1.2. Let $\iota : L \longrightarrow M^{2n}$ be an immersion into a symplectic manifold $\left(M^{2n}, \Omega\right)$. We say that L is a **Lagrangian submanifold** if the dimension of L is n and $\iota^*\Omega = 0$.

Theorem 1.1.3 (Weistein). *If L is a Lagrangian submanifold of a symplectic manifold (M, Ω), then there exists a neighborhood of L in M that is symplectomorphic to a neighborhood of the zero section in T^*L.*

Theorem 1.1.4. *Let X be a symplectic vector field on $(M, \Omega), g$ an associated metric, and J the corresponding almost complex structure. Then $X^i = J^{ik}\theta_k$ for some closed 1-form θ. Conversely, given a closed 1-form $\theta, X^i = J^{ik}\theta_k$ defines a symplectic vector field.*

Corollary 1.1.5. *For $f \in C^\infty(M), J\nabla f$ is symplectic, where ∇f is the gradient of f. Conversely, given X symplectic, X is locally $J\nabla f$.*

Theorem 1.1.6 (Hatakeyama, 1966). *The group of symplectomorphisms acts transitively on a compact symplectic manifold (M, Ω).*

§ 1.1.1 Examples

Example 1.1.7. Let M be a differentiable manifold. Then its cotangent bundle T^*M has a natural symplectic structure. For $z \in T^*M$ and V in the tangent space of T^*M at z, T_zT^*M, define a 1 -form β, often called the **Liouville form**, by $\beta(V)_z = z\left(\pi_* V\right)$, where $\pi : T^*M \longrightarrow M$ is the projection map. If $x^1, \ldots, x^n$ are local coordinates on M, then $q^i = x^i \circ \pi$ together with fiber coordinates $p^1, \ldots, p^n$ give local coordinates on T^*M. In these coordinates β has the local expression $\sum_{i=1}^n p^i dq^i$. The natural symplectic structure on T^*M is given by $\Omega = -d\beta$.

§ 1.2 Contact manifolds

Definition 1.2.1. By a **contact manifold** we mean a C^∞ manifold M^{2n+1} together with a 1 -form η such that

$$\eta \wedge (d\eta)^n \neq 0. \tag{1.1}$$

Remark 1.2.1. 1) In particular, $\eta \wedge (d\eta)^n \neq 0$ is a volume element on M, so that a contact manifold is orientable.

2) we have a global vector field ξ satisfying

$$d\eta(\xi, X) = 0, \quad \eta(\xi) = 1. \tag{1.2}$$

ξ is called the **characteristic vector field** or **Reeb vector field** of the contact structure η.

3) By formula $L_\xi = d\circ(\xi\lrcorner) + (\xi\lrcorner)\circ d$, we have immediately that

$$L_\xi \eta = 0, \quad L_\xi d\eta = 0. \tag{1.3}$$

Definition 1.2.2. We denote by $\mathcal{D}$ the **contact distribution** or subbundle defined by the subspaces $\mathcal{D}_m = \{X \in T_m M \,:\, \eta(X) = 0\}$.

Remark 1.2.2. Roughly speaking, the meaning of the contact condition, $\eta \wedge (d\eta)^n \neq 0$, is that the contact subbundle is as far from being integrable as possible; in particular, $\mathcal{D}$ rotates as one moves around on the manifold. For a subbundle defined by a 1 -form η to be integrable it is necessary and sufficient that

$$\eta \wedge (d\eta) \equiv 0. \tag{1.4}$$

Definition 1.2.3. A contact structure is **regular** if ξ is **regular** as a vector field, that is, every point of the manifold has a neighborhood such that any integral curve of the vector field passing through the neighborhood passes through only once.

Remark 1.2.3. Two well-known examples of nonregular vector fields on surfaces are the irrational flow on a torus and the flow around a Möbius band.

Theorem 1.2.4 (Darboux coordinate system). *About each point of a contact manifold (M^{2n+1}, η) there exist local coordinates $(x^1, \ldots, x^n, y^1, \ldots, y^n, z)$ with respect to which*

$$\eta = dz - \sum_{i=1}^{n} y^i dx^i. \tag{1.5}$$

Remark 1.2.4. There are many similar results in this spirit.

Definition 1.2.5. A diffeomorphism f of M^{2n+1} or between open subsets of $\mathbb{R}^{2n+1}$ with the contact structure of the Darboux form of the above theorem is called a **contact transformation** if

$$f^*\eta = \tau\eta \tag{1.6}$$

for some nonvanishing function τ on the domain of f. If $\tau \equiv 1$, f is called a **strict contact transformation**.

Definition 1.2.6. There is also the notion of a **contact structure in the wider sense**, often called simply a "contact structure" by many authors, which can be defined in a number of ways. For example, a contact manifold in the wider sense is a manifold with a differentiable structure modeled on the pseudogroup of contact transformations on $\mathbb{R}^{2n+1}$.

Theorem 1.2.7. *Let M^{2n+1} be a contact manifold in the wider sense. If n is odd, then M^{2n+1} is orientable.*

Theorem 1.2.8. *Let M^{2n+1} be a contact manifold in the wider sense. If n is even and M^{2n+1} is orientable, then M^{2n+1} is a contact manifold.*

Theorem 1.2.9. *Let M^{2n+1}be a contact manifold in the wider sense which is not a contact manifold in the restricted sense. Then its 2-fold covering manifold is a contact manifold in the restricted sense.*

Theorem 1.2.10 (Boothby-Wang fibration). *Let $\left(M^{2n+1}, \eta'\right)$ be a compact regular contact manifold. Then there exists a contact form $\eta = \tau\eta'$ for some nonvanishing function τ whose characteristic vector field ξ generates a free effective S^1 action on M^{2n+1}. Moreover, M^{2n+1} is the bundle space of a principal circle bundle $\pi : M^{2n+1} \longrightarrow M^{2n}$ over a symplectic manifold M^{2n} whose symplectic form Ω determines an integral cocycle on M^{2n} and η is a connection form on the bundle with curvature form $d\eta = \pi^*\Omega$.*

Definition 1.2.11 (Other notions). periodic point, planar, almost regular/quasi-regular, R-contact manifold, etc..

§ 1.2.1 Examples

Example 1.2.12. The cotangent sphere bundle and the tangent sphere bundle of a Riemannian manifold are contact manifolds.

Example 1.2.13. Let M be an n-dimensional manifold and T^*M its cotangent bundle. As in the previous example we can define a 1 -form β by the local expression $\beta = \sum_{i=1}^n p^i dq^i$. Let $M^{2n+1} = T^*M \times \mathbb{R}, t$ the coordinate on $\mathbb{R}$, and $\mu : M^{2n+1} \longrightarrow T^*M$ the projection to the first factor. Then $\eta = dt - \mu^*\beta$ is a contact form on M^{2n+1}.

§ 1.3 Associated Metrics

Definition 1.3.1. An **almost complex structure** is a tensor field J of type $(1,1)$ such that $J^2 = -I$. A **Hermitian metric** on an almost complex manifold (M, J) is a Riemannian metric that is invariant by J, i.e.,

$$g(JX, JY) = g(X, Y). \tag{1.7}$$

Remark 1.3.1. Note also that every almost complex manifold admits a Hermitian metric, for if k is any Riemannian metric, then g defined by

$$g(X,Y) = k(X,Y) + k(JX,JY) \tag{1.8}$$

is Hermitian.

Definition 1.3.2. Noting that J is negative self-adjoint with respect to g, i.e., $g(X,JY) = -g(JX,Y)$, therefore

$$\Omega(X,Y) = g(X,JY) \tag{1.9}$$

defines a 2 -form called the **fundamental 2-form** of the **almost Hermitian structure** (M,J,g). If

$$d\Omega = 0, \tag{1.10}$$

the structure is **almost Kähler**. If M is a complex manifold and J the corresponding almost complex structure, we say that (M,J,g) is a **Hermitian manifold**.

Definition 1.3.3. Given (M,J,g) we can construct a particular local orthonormal basis as follows. Let $\mathcal{U}$ be a coordinate neighborhood on M and X_1 any unit vector field on $\mathcal{U}$. Let $X_{1^*} = JX_1$. Now choose a unit vector field X_2 orthogonal to both X_1 and X_{1^*}. Then JX_2 is also orthogonal to X_1 and X_{1^*}. Continuing in this manner we have a local orthonormal basis of the form $\left\{X_1,\dots,X_n,JX_1,\dots,JX_n\right\}$. Such a basis is called a J**-basis.**

Remark 1.3.2. An almost complex (almost Hermitian) structure on M can be thought of as a reduction of the structural group to $U(n)$.

Remark 1.3.3. The structural group of a symplectic manifold is reducible to $U(n)$ and that of a contact manifold to $U(n)\times 1$ (Chern [1953]).

Definition 1.3.4. For an odd-dimensional manifold M^{2n+1}, J. Gray [1959] defined **an almost contact structure** as a reduction of the structural group to $U(n)\times 1$. In terms of structure tensors we say that M^{2n+1} has an almost contact structure or sometimes (ϕ,ξ,η)-**structure** if it admits a tensor field ϕ of type $(1,1)$, a vector field ξ, and a 1 -form η satisfying

$$\phi^2 = -I + \eta\otimes\xi, \quad \eta(\xi) = 1. \tag{1.11}$$

Theorem 1.3.5. *Suppose M^{2n+1} has a (ϕ,ξ,η)-structure. Then*

$$\phi\xi = 0, \quad \eta\circ\phi = 0. \tag{1.12}$$

Moreover, the endomorphism ϕ has rank $2n$.

Definition 1.3.6. If a manifold M^{2n+1} with a (ϕ,ξ,η)-structure admits a Riemannian metric g such that

$$g(\phi X,\phi Y) = g(X,Y) - \eta(X)\eta(Y), \tag{1.13}$$

we say that M^{2n+1} has an **almost contact metric structure** and g is called a **compatible metric**.

Remark 1.3.4. For any $X \in \mathcal{D}$, let $Y = \xi$ in (1.13), we get $g(\phi(X), \xi) = 0$, thus

$$\phi(\mathcal{D}) \subset \mathcal{D}. \tag{1.14}$$

Remark 1.3.5. As in the almost Hermitian case, we define a 2 -form, called the fundamental 2-form of the almost contact metric structure, by

$$\Phi(X, Y) = g(X, \phi Y). \tag{1.15}$$

Remark 1.3.6. Also as in the case of an almost complex structure, the existence of the compatible metric is easy. For if k' is any metric, first set $k(X, Y) = k'\left(\phi^2 X, \phi^2 Y\right) + \eta(X)\eta(Y)$; then $\eta(X) = k(X, \xi)$. Now define g by

$$g(X, Y) = \frac{1}{2}(k(X, Y) + k(\phi X, \phi Y) + \eta(X)\eta(Y)). \tag{1.16}$$

Remark 1.3.7. Setting $Y = \xi$ we have immediately that

$$\eta(X) = g(X, \xi). \tag{1.17}$$

Remark 1.3.8. We can obtain a local orthonormal basis $\left\{X_i, X_{i^*} = \phi X_i, \xi\right\}$, called a **$\phi$-basis**.

Remark 1.3.9. Given a manifold M^{2n+1} with a (ϕ, ξ, η)-structure, let g be a compatible metric, then proceeding as in the almost complex case, we see that the structural group of M^{2n+1} is reducible to $U(n) \times 1$.

Theorem 1.3.7 (Existence of associated metrics 1). *Let $\left(M^{2n}, \Omega\right)$ be a symplectic manifold. Then there exist a Riemannian metric g and an almost complex structure J such that*

$$g(X, JY) = \Omega(X, Y).$$

Definition 1.3.8. In particular, given a symplectic manifold $\left(M^{2n}, \Omega\right)$, we say that a Riemannian metric g is an **associated metric** if there exists an almost complex structure J such that

$$g(X, JY) = \Omega(X, Y).$$

Theorem 1.3.9 (Existence of associated metrics 2). *Let $\left(M^{2n+1}, \eta\right)$ be a contact manifold and ξ its characteristic vector field. Then there exists an almost contact metric structure such that*

$$g(X, \phi Y) = d\eta(X, Y).$$

Definition 1.3.10. As in the symplectic case, given a contact manifold $\left(M^{2n+1}, \eta\right)$, we say that a Riemannian metric g is an **associated metric** if there exists an almost contact metric structure such that

$$g(X, \phi Y) = d\eta(X, Y).$$

In this case we also speak of a **contact metric structure**; other authors often use the phrase contact Riemannian structure.

Definition 1.3.11 (Alternative definition). Working strictly with structure tensors, one may avoid the polarization process by defining an associated metric as follows: Given a contact manifold $\left(M^{2n+1}, \eta\right)$ with characteristic vector field ξ, a Riemannian metric g is an associated metric if, first of all,

$$\eta(X)=g(X, \xi)$$

and secondly, there exists a tensor field ϕ of type $(1,1)$ such that

$$\phi^2=-I+\eta \otimes \xi, \quad d \eta(X, Y)=g(X, \phi Y).$$

Proof. Only need to verify

$$g(X, \phi Y)+g(\phi X, Y)=0, X, Y \in \mathcal{D}. \tag{1.18}$$

But it follows immediately by symmetry of g and antisymmetry of $d\eta$. □

Remark 1.3.10. Caution that it is possible to have a contact manifold $\left(M^{2n+1}, \eta\right)$ with characteristic vector field ξ and an almost contact metric structure (ϕ, ξ, η, g), same ξ and η, without $g(X, \phi Y)=d \eta(X, Y)$.

Remark 1.3.11. On the other hand, it is possible for a Riemannian metric g to be an associated metric for more than one symplectic structure. For example, on a hyper-Kähler manifold one has, by definition, three independent global Kähler structures $\left(J_a, g\right), a=1,2,3$, satisfying $J_1 J_2+J_2 J_1=0$, $J_3=J_1 J_2$. Thus the three fundamental 2 -forms give three symplectic structures with g an associated metric for each of them.

Theorem 1.3.12. *On a contact metric manifold the integral curves of ξ are geodesics.*

Proof. For a contact metric structure we have

$$0=\left(L_{\xi} \eta\right)(X)=\xi g(X, \xi)-g\left(\nabla_{\xi} X-\nabla_X \xi, \xi\right)=g\left(X, \nabla_{\xi} \xi\right),$$

so the integral curves of ξ are geodesics. □

Remark 1.3.12. The set $\mathcal{A}$ of all associated metrics forms a infinite-dimensional sapce.

Theorem 1.3.13. *Let $\left(M^{2n}, \Omega\right)$ be a symplectic manifold (resp. $\left(M^{2n+1}, \eta\right)$ a contact manifold) and g an associated metric. Then*

$$d V=\frac{(-1)^n}{2^n n!} \Omega^n\left(\text{resp. } d V=\frac{(-1)^n}{2^n n!} \eta \wedge(d \eta)^n\right). \tag{1.19}$$

Remark 1.3.13. Not sure if it's correct up to some factors.

Definition 1.3.14. $\mathcal{A}$: the set of all associated metrics
$\mathcal{N}$: the set of all metrics with the same volume element
$\mathcal{M}$: the set of all metrics

Definition 1.3.15. For symmetric tensor fields S and T of type $(0,2)$ we define a Riemannian metric $(\cdot,\cdot)$ by

$$(S,T)_g = \int_M S_{ij}T_{kl}g^{ik}g^{jl}dV_g. \tag{1.20}$$

Remark 1.3.14. One can think of metrics with the same total volume on an n-dimensional manifold as being at a fixed distance from the zero tensor.

Proof. Consider the path $tg, t \in [0,1]$, from the zero tensor to the metric g; then since g may be considered as the tangent to the path at each point, we have the following entertaining computation:

$$|g| \equiv \int_0^1 (g,g)_{tg}^{1/2}dt = \int_0^1 \left(\frac{n}{t^2}\int_M \sqrt{t^n \det g}\,dx^1\cdots dx^n\right)^{1/2} dt = 4\sqrt{\frac{\mathrm{vol}_g M}{n}}.$$

□

Proposition 1.3.16. *On a compact manifold M the set $\mathcal{M}$ of all Riemannian metrics may be given a Riemannian metric (Ebin [1970]): The tangent space, $T_g\mathcal{M}$, of $\mathcal{M}$ at a metric g is the space of symmetric tensor fields of type* $(0,2)$.

The tangent space $T_g\mathcal{A}$ of $\mathcal{A}$ at g is the set of all symmetric tensor fields that anticommute with J (ϕ and annihilate ξ).

Theorem 1.3.17. *Let (M,Ω) be a symplectic manifold, or respectively (M,η) a contact manifold, and f a diffeomorphism satisfying $f^*\Omega = \Omega$, respectively $f^*\eta = \eta$. Then for any associated metric g, f^*g is also an associated metric.*

Lemma 1.3.18. *Let (M,g) be a compact orientable Riemannian manifold and for a vector field V let $v(X) = g(V,X)$. Then $\int_M V^i\theta_i dV_g = 0$ for every closed 1-form θ if and only if $v = \delta\Psi$ for some 2 -form Ψ. $\int_M v_i X^i dV_g = 0$ for every vector field X if and only if $v = 0$.*

§ 1.4 Examples of almost contact metric manifolds

Example 1.4.1. $\mathbb{R}^{2n+1}(-3)$. It's a contact metric structure on it.

Example 1.4.2. $M^{2n+1} \subset \tilde{M}^{2n+2}$. Every C^∞ orientable hypersurface of an almost complex manifold has an almost contact structure.

In general, one cannot expect the induced almost contact metric structure to be a contact metric structure.

Theorem 1.4.3. *Let M^{2n+1} be a hypersurface of a Kähler manifold $\tilde{M}^{2n+2}$, (ϕ,ξ,η,g) its induced almost contact metric structure and A its Weingarten map. Then (ϕ,ξ,η,g) is a contact metric structure if and only if*

$$A\phi + \phi A = -2\phi. \tag{1.21}$$

Example 1.4.4. $S^5 \subset S^6$.

Example 1.4.5 (The Boothby–Wang fibration). It has a **K-contact structure**.

Definition 1.4.6. A contact metric structure for which the characteristic vector field is a Killing vector field is called a **K-contact structure**.

Theorem 1.4.7. *Let $\pi : M^{2n+1} \to M^{2n}$ be the Boothby-Wang fibration of a compact regular contact manifold M^{2n+1}. Then the first Betti numbers of M^{2n+1} and M^{2n} are equal.*

Theorem 1.4.8. *No torus T^{2n+1} can carry a regular contact structure.*

Example 1.4.9. $M^{2n} \times \mathbb{R}$. It is an almost contact metric structure that is certainly not a contact metric structure.

Example 1.4.10 (Parallelizable manifolds). In particular, any odd-dimensional Lie group carries an almost contact structure.

Definition 1.4.11. A smooth manifold with or without boundary is said to be **parallelizable** if it admits a smooth global frame.

Theorem 1.4.12. *A smooth vector bundle is smoothly trivial if and only if it admits a smooth global frame.*

§ 1.5 Integral Submanifolds and Contact Transformations

§ 1.5.1 Integral submanifolds

Let M^{2n+1} be a contact manifold with contact form η. We have seen that $\eta = 0$ defines a $2n$-dimensional subbundle $\mathcal{D}$ called the **contact distribution** or **subbundle** and that since $\eta \wedge (d\eta)^n \neq 0, \mathcal{D}$ is nonintegrable.

Definition 1.5.1. A submanifold M^r of M^{2n+1} is called an **integral submanifold** if $\eta(X) = 0$ for every tangent vector X. It is clear that for any pair of tangent vector fields we have

$$d\eta(X,Y) = \frac{1}{2}(X\eta(Y) - Y\eta(X) - \eta([X,Y]) = 0.$$

Then in terms of associated metrics, $g(X, \phi Y) = 0$ and for this reason integral submanifolds are often called **C-totally real submanifolds.** In particular, ϕ maps tangent vectors to normal vectors; also, since ξ is a normal vector, the dimension r can be at most n. On the other hand, by the Darboux theorem, it's easy to see there exist an n-dim integral submanifold.

Theorem 1.5.2. *Let M^{2n+1} be a contact manifold with contact form η. Then there exist integral submanifolds of the contact subbundle $\mathcal{D}$ of dimension n but of no higher dimension.*

Theorem 1.5.3 (Sasaki, 1964). *Let $\left(x^i, y^i, z\right)$, $i = 1, \ldots, n$, be local coordinates about a point $m = \left(x_0^i, y_0^i, z_0\right)$ such that $\eta = dz - \sum_{i=1}^n y^i dx^i$ on the coordinate neighborhood. In order that r linearly independent vectors X_λ, $\lambda = 1, \ldots, r \le n$, at m with components $\left(a_\lambda^i, b_\lambda^i, c_\lambda\right)$ be tangent to an r dimensional integral submanifold it is necessary and sufficient that $\eta\left(X_\lambda\right) = 0$ and $d\eta\left(X_\lambda, Y_\mu\right) = 0$, that is, $c_\lambda = \sum_i y_0^i a_\lambda^i$ and $\sum_i a_\lambda^i b_\mu^i = \sum_i a_\mu^i b_\lambda^i$.*

As in the symplectic case we note the abundance of integral submanifolds of $\mathcal{D}$.

Theorem 1.5.4 (Sasaki, 1964). *Given a vector $X \in \mathcal{D}$ at $m \in M^{2n+1}$ and any $r, 1 \le r \le n$, there exists an r-dimensional integral submanifold M^r of $\mathcal{D}$ through m with X tangent to M^r.*

Similar to Weistein theorem, we have

Theorem 1.5.5 (Lychagin, 1977). *If L is an n-dimensional integral submanifold of a contact manifold $\left(M^{2n+1}, \eta\right)$, then there exist an open neighborhood $\mathcal{U}$ of L in M^{2n+1}, an open neighborhood $\mathcal{V}$ of the zero section in $T^*L \times \mathbb{R}$, and a diffeomorphism $f : \mathcal{U} \longrightarrow \mathcal{V}$ such that $f|_L$ is the identity on L and $f^*(\beta - dt) = \eta$.*

§ 1.5.2 Contact transformations

Recall that a diffeomorphism f of M^{2n+1} is a contact transformation if $f^*\eta = \tau\eta$ for some nonvanishing function τ and that f is a strict contact transformation if $\tau \equiv 1$. Clearly $f^*d\eta = d\tau \wedge \eta + \tau d\eta$, so if $d\eta$ is invariant, $(\tau - 1)d\eta = -d\tau \wedge \eta$ and hence $(\tau - 1)\eta \wedge d\eta = 0$, giving $\tau \equiv 1$. Thus f is strict if and only if $d\eta$ is invariant.

Theorem 1.5.6. *f is a contact transformation if and only if $X \in \mathcal{D}$ implies $f_*X \in \mathcal{D}$.*

Theorem 1.5.7. *A diffeomorphism f is a contact transformation if and only if f maps r-dimensional integral submanifolds to r-dimensional integral submanifolds.*

Definition 1.5.8. If $L_X\eta = \sigma\eta$ for some function σ, X is called an **infinitesimal contact transformation**.If $\sigma = 0$, we say that X is a **strict infinitesimal contact transformation**.

Theorem 1.5.9. *A vector field X is an infinitesimal contact transformation if and only if there exists a function f on M^{2n+1} such that*

$$X = -\frac{1}{2}\phi\nabla f + f\xi. \tag{1.22}$$

Proof. For sufficiency, we have

$$L_X\eta = (\xi f)\eta. \tag{1.23}$$

... □

Corollary 1.5.10. *X is strict if and only if $\xi f = 0$.*

As in the symplectic case we have the following theorem of Hatakeyama on the transitivity of the group of contact transformations.

Theorem 1.5.11 (Hatakeyama, 1966). *Let M^{2n+1} be a compact contact manifold. Then for any two points p, q, there exists a contact transformation mapping p to q. If M^{2n+1} is regular, there exists a strict contact transformation mapping p to q.*

§ 1.6 Sasakian and Cosymplectic Manifolds

Let M^{2n+1} be an almost contact manifold with structure tensors (ϕ, ξ, η) and consider the manifold $M^{2n+1} \times \mathbb{R}$. We denote a vector field on $M^{2n+1} \times \mathbb{R}$ by $(X, f\frac{d}{dt})$, where X is tangent to M^{2n+1}, t the coordinate on $\mathbb{R}$, and f a C^∞ function on $M^{2n+1} \times \mathbb{R}$. Define an almost complex structure J on $M^{2n+1} \times \mathbb{R}$ by

$$J\left(X, f\frac{d}{dt}\right) = \left(\phi X - f\xi, \eta(X)\frac{d}{dt}\right)$$

that $J^2 = -I$ is easy to check. If now J is integrable, we say that the almost contact structure (ϕ, ξ, η) is **normal** (Sasaki and Hatakeyama [1961]).

Remark 1.6.1. Compare to the "regular".

Definition 1.6.1. We are thus led to define four tensors $N^{(1)}, N^{(2)}, N^{(3)}, N^{(4)}$ by

$$\begin{aligned} N^{(1)}(X,Y) &= [\phi,\phi](X,Y) + 2d\eta(X,Y)\xi, \\ N^{(2)}(X,Y) &= \left(L_{\phi X}\eta\right)(Y) - \left(L_{\phi Y}\eta\right)(X), \\ N^{(3)} &= \left(L_\xi \phi\right) X \\ N^{(4)} &= \left(L_\xi \eta\right)(X). \end{aligned} \tag{1.24}$$

Theorem 1.6.2. *For an almost contact structure (ϕ, ξ, η) the vanishing of $N^{(1)}$ implies the vanishing of $N^{(2)}, N^{(3)}$ and $N^{(4)}$.*

Theorem 1.6.3. *For a contact structure (ϕ, ξ, η, g), $N^{(2)}$ and $N^{(4)}$ vanish. Moreover, $N^{(3)}$ vanishes if and only if ξ is a Killing vector field.*

Lemma 1.6.4. *For an almost contact metric structure (ϕ, ξ, η, g), the covariant derivative of ϕ is given by*

$$\begin{aligned} 2g\left(\left(\nabla_X \phi\right)Y, Z\right) =& 3d\Phi(X, \phi Y, \phi Z) - 3d\Phi(X, Y, Z) + g\left(N^{(1)}(Y,Z), \phi X\right) \\ &+ N^{(2)}(Y,Z)\eta(X) + 2d\eta(\phi Y, X)\eta(Z) \\ &- 2d\eta(\phi Z, X)\eta(Y). \end{aligned} \tag{1.25}$$

Definition 1.6.5. Define a tensor

$$h := \frac{1}{2}\left(L_\xi \phi\right) X = \frac{1}{2} N^{(3)}. \tag{1.26}$$

Remark 1.6.2. The first property to note is immediate, namely $h\xi = 0$.

Lemma 1.6.6. *On a contact metric manifold h is a symmetric operator,*

$$\nabla_X \xi = -\phi X - \phi h X. \tag{1.27}$$

h anticommutes with ϕ, and $\operatorname{tr} h = 0$.

Corollary 1.6.7. *On a contact metric manifold $\delta\eta = 0$.*

§ 1.6.1 Definition of Sasakian manifolds

Definition 1.6.8. A Sasakian manifold is a normal contact metric manifold.

Theorem 1.6.9. *An almost contact metric structure (ϕ, ξ, η, g) is Sasakian if and only if*

$$\left(\nabla_X \phi\right) Y = g(X, Y)\xi - \eta(Y)X. \tag{1.28}$$

Corollary 1.6.10. *A Sasakian manifold is K-contact.*

Remark 1.6.3. In dimension 3 the converse is true, Corollary 6.5. For K-contact structures that are not Sasakian, see Example 6.7.2.

Let's draw two diagrams to make a conclusion:

$$\begin{tikzcd}
\text{Hermitian} \arrow[r, "d\Omega = 0"] & \text{Kaehler} \\
\text{almost Hermitian} \arrow[u, "{[J,J] = 0}"] \arrow[r, "d\Omega = 0"] \arrow[ur, "\nabla J = 0"] & \text{almost Kaehler} \arrow[u, "{[J,J] = 0}"']
\end{tikzcd}
\qquad
\begin{tikzcd}
\text{normal almost contact metric} \arrow[r, "\Phi = d\eta"] & \text{Sasakian} \\
\text{almost contact metric} \arrow[u, "N^{(1)} = 0"] \arrow[r, "\Phi = d\eta"] \arrow[ur, "(1.28)"] & \text{contact metric} \arrow[u, "N^{(1)} = 0"']
\end{tikzcd}$$

§ 1.6.2 More about Sasakian manifolds

Recall that a **contact metric structure** on an odd-dimensional manifold N^{2n+1} is given by (φ, ξ, η, g), where φ is a tensor field of type $(1,1)$ on N, ξ is a vector field, η is an 1-form and g is a Riemannian metric such that

$$\varphi^2 = -I + \eta \otimes \xi, \quad \eta(\xi) = 1 \tag{1.29}$$

and

$$g(\varphi X, \varphi Y) = g(X, Y) - \eta(X)\eta(Y), \quad g(X, \varphi Y) = d\eta(X, Y), \quad \forall X, Y \in C(TN). \tag{1.30}$$

Remark 1.6.4. Recall that [5], from (1.29), one can deduce

$$\varphi\xi = 0, \quad \eta \circ \varphi = 0.$$

Example 1.6.11. Strictly pseudoconvex CR manifolds are therefore contact Riemannian manifolds, in a natural way.

Theorem 1.6.12. *Let $\left(M, T_{1,0}(M), \theta\right)$ be a pseudo-Hermitian manifold. Then the Webster metric g_θ is Sasakian if and only if the Tanaka-Webster connection of $\left(M, T_{1,0}(M), \theta\right)$ has vanishing pseudo-Hermitian torsion, that is, $\tau = 0$.*

Recall that a contact metric manifold $(N, \varphi, \xi, \eta, g)$ is called **Sasakian** if it is normal, i.e.

$$N_\varphi + 2d\eta \otimes \xi = 0,$$

where

$$N_\varphi(X, Y) = [\varphi X, \varphi Y] - \varphi[\varphi X, Y] - \varphi[X, \varphi Y] + \varphi^2[X, Y], \quad \forall X, Y \in C(TN), \tag{1.31}$$

is the **Nijenhuis tensor** field of φ, or, equivalently, if

$$\left(\nabla_X \varphi\right)(Y) = g(X, Y)\xi - \eta(Y)X, \quad \forall X, Y \in C(TN). \tag{1.32}$$

Remark 1.6.5. We note that from the above formula it follows

$$\nabla_X \xi = -\varphi X. \tag{1.33}$$

§ 1.6.3 Connections and curvatures

Theorem 1.6.13. *Let $\left(M, T_{1,0}(M), \theta\right)$ be a pseudo-Hermitian manifold and g_θ the Webster metric of $\left(M, T_{1,0}(M), \theta\right)$. Then there exists a unique linear connection ∇ on M, called the* ***Tanaka-Webster*** *connection, such that:*

(1) the Levi distribution $H(M)$ is parallel with respect to ∇;

(2) $\nabla g_\theta = 0, \nabla J = 0, \nabla \theta = 0$ (hence $\nabla T = 0$);

(3) the torsion T_∇ of ∇ satisfies $T_\nabla(X, Y) = -2\Omega(X, Y)T$ and $T_\nabla(T, JX) = -JT_\nabla(T, X)$, for any $X, Y \in H(M)$.

We denote by ∇^θ the Levi-Civita connection of $\left(M, g_\theta\right)$. One can see that the Levi-Civita connection ∇^θ is related to the Tanaka-Webster connection ∇ by

$$\nabla^\theta = \nabla + (\Omega - A) \otimes T + \tau \otimes \theta + 2\theta \odot J, \tag{1.34}$$

where $2(\theta \odot J)(X, Y) = \theta(X)JY + \theta(Y)JX$. Thus, we have $\nabla^\theta_X T = \tau(X) + JX$. In particular,

$$\nabla^\theta_T T = 0. \tag{1.35}$$

If $X, Y \in H(M)$, then

$$\nabla^\theta_X Y = \nabla_X Y + [\Omega(X, Y) - A(X, Y)]T. \tag{1.36}$$

Since $\nabla^\theta - \nabla$ is a $(1, 2)$ tensor field on M and $\nabla^\theta_T T = \nabla_T T = 0$, we can define a vector field V on M given by

$$V = \operatorname{trace}_{g_\theta}\left(\nabla^\theta - \nabla\right) = \operatorname{trace}_{G_\theta}\left(\nabla^\theta - \nabla\right). \tag{1.37}$$

Actually we have

$$V = -\operatorname{trace}_{g_\theta}(\tau)T = 0 \tag{1.38}$$

Definition 1.6.14. Let $(N, \varphi, \xi, \eta, g)$ be a Sasakian manifold. The sectional curvature of a 2 -plane generated by X and φX, where X is an unit vector orthogonal to ξ, is called the φ-**sectional curvature** determined by X. If the φ-sectional curvature is a constant c, then $(N, \varphi, \xi, \eta, g)$ is called a **Sasakian space form** and it is denoted by $N(c)$. The curvature tensor field of a Sasakian space form $N(c)$ is given by

$$\begin{aligned} R(X,Y)Z =& \frac{c+3}{4}\{g(Z,Y)X - g(Z,X)Y\} + \frac{c-1}{4}\{\eta(Z)\eta(X)Y - \\ & - \eta(Z)\eta(Y)X + g(Z,X)\eta(Y)\xi - g(Z,Y)\eta(X)\xi + \\ & + g(Z,\varphi Y)\varphi X - g(Z,\varphi X)\varphi Y + 2g(X,\varphi Y)\varphi Z\} \end{aligned} \tag{1.39}$$

Remark 1.6.6 (cf. [4]). The φ-sectional curvature determines the curvature tensor R completely.

Remark 1.6.7. A direct consequence of (1.34) is that, for Sasakian manifolds, we have(substitute $\tau = 0, A = 0$)

$$K(X, JX) = K^{\theta}(X, JX) + 3, \forall X \in \mathcal{H}(M), \tag{1.40}$$

and

$$K(X, T) = K^{\theta}(X, T) - 1 = 0, \forall X \in \mathcal{H}(M), \tag{1.41}$$

where K is sectional curvature wrt TW connection, while K^{θ} is the sectional curvature wrt LC connection.

Theorem 1.6.15 (cf. [3]). *Let $M(c)$ be a complete, simply connected Sasakian manifold with constant ϕ-sectional curvature c. Then $M(c)$ belongs one of the three families of examples listed below.*

To begin, let (ϕ, ξ, η, g) be a contact metric structure and recall the notion of a $\mathcal{D}$-**homothetic deformation**:

$$\bar{\eta} = a\eta, \quad \bar{\xi} = \frac{1}{a}\xi, \quad \bar{\phi} = \phi, \quad \bar{g} = ag + a(a-1)\eta \otimes \eta, \tag{1.42}$$

where a is a positive constant. The deformed structure $(\bar{\phi}, \bar{\xi}, \bar{\eta}, \bar{g})$ is again a contact metric structure, and it enjoys many of the properties of the original structure. In particular, if (ϕ, ξ, η, g) is Sasakian, so is $(\bar{\phi}, \bar{\xi}, \bar{\eta}, \bar{g})$; if $M(c)$ is a Sasakian space form, then deforming the structure, we obtain the Sasakian space form $M(\bar{c})$ where

$$\bar{c} = \frac{c+3}{a} - 3$$

(see Tanno [1968], [1969] for details). We will show that there exist Sasakian space forms $M(c)$ for every value of c.

Remark 1.6.8. (1.42) is natural from my point of view: we want to make a conformal transformation on $\mathcal{D}$ or ,more precisely, homothetic deformation, so we hope when restricting to $\mathcal{D}$: $\bar{g} = ag$, so we may assume $\bar{g} = ag + b\eta \otimes \eta$. But from the 2nd of (1.30), we are suggested to let $\bar{\eta} = a\eta$ so $\bar{\xi}$ follows its step. Now put the above transformations to the 1st of (1.30) (testing $\bar{\xi}$), we get $b = a^2 - a$.

Example 1.6.16 **($\mathbb{S}^{2n+1}, c > -3$).** Let $\mathbb{S}^{2n+1} = \left\{z \in \mathbb{C}^{n+1} : |z| = 1\right\}$ be the unit $(2n+1)$-dimensional Euclidean sphere. Consider the following structure tensor fields on $\mathbb{S}^{2n+1}$: the standard metric field g_0, the vector field $\xi_0(z) = -\mathcal{J} z, z \in \mathbb{S}^{2n+1}$, where $\mathcal{J}$ is the usual almost complex structure on $\mathbb{C}^{n+1}$ defined by

$$\mathcal{J} z = \left(-y^1, \ldots, -y^{n+1}, x^1, \ldots, x^{n+1}\right),$$

for $z = \left(x^1, \ldots, x^{n+1}, y^1, \ldots, y^{n+1}\right)$, and $\varphi_0 = s \circ \mathcal{J}$, where $s : T_z\mathbb{C}^{n+1} \to T_z\mathbb{S}^{2n+1}$ denotes the orthogonal projection. Then g_0 and ξ_0 detemine η_0 and φ_0 by

$$\eta_0 = g_0(\xi_0, \cdot), \quad d\eta_0(X, Y) = 2g(X, \varphi_0 Y).$$

Equipped with these tensors, $\mathbb{S}^{2n+1}$ becomes a Sasakian space form with the φ_0-sectional curvature equal to 1 , denoted by $\mathbb{S}^{2n+1}(1)$. Now, consider the deformed structure on $\mathbb{S}^{2n+1}$

$$\eta = a\eta_0, \quad \xi = \frac{1}{a}\xi_0, \quad \varphi = \varphi_0, \quad g = ag_0 + a(a-1)\eta_0 \otimes \eta_0$$

where a is a positive constant. The structure (φ, ξ, η, g) is still a Sasakian structure and $\left(\mathbb{S}^{2n+1}, \varphi, \xi, \eta, g\right)$ is a Sasakian space form with constant φ-sectional curvature $c = \frac{4}{a} - 3$, $c > -3$, denoted by $\mathbb{S}^{2n+1}(c)$.

Example 1.6.17 **($\mathbb{R}^{2n+1}, c = -3$).** The Euclidean space $\mathbb{R}^{2n+1}\left(x^1, \ldots, x^n, y^1, \ldots, y^n, z\right)$ is a Sasaki manifold with Sasaki structure (ξ, η, ϕ, g) as follows:

$$\xi = 2\frac{\partial}{\partial z}, \quad \eta = \frac{1}{2}\left(dz - \sum_{i=1}^{n} y^i dx^i\right), \quad g = \eta \otimes \eta + \frac{1}{4}\sum_{i=1}^{n}\left(\left(dx^i\right)^2 + \left(dy^i\right)^2\right),$$

$$\phi\left(\frac{\partial}{\partial x^i}\right) = -\frac{\partial}{\partial y^i}, \quad \phi\left(\frac{\partial}{\partial y^i}\right) = \frac{\partial}{\partial x^i} + y^i\frac{\partial}{\partial z}, \quad \phi\left(\frac{\partial}{\partial z}\right) = 0.$$

Example 1.6.18 **($B^n \times \mathbb{R}, c < -3$).** Let B^n be a simply connected bounded domain in $\mathbb{C}^n$ and (J, G) a Kähler structure with constant holomorphic sectional curvature $k < 0$. For such a structure, the fundamental 2 -form Ω of the Kähler structure is exact and hence $\Omega = d\omega$ for some real analytic 1 -form ω. Now on $B^n \times \mathbb{R}$ let π denote the projection onto B^n and t the coordinate on $\mathbb{R}$. Then $B^n \times \mathbb{R}$ with $\eta = \pi^*\omega + dt$ and $g = \pi^*G + \eta \otimes \eta$ is a Sasakian manifold. Regarding η as a connection form on $B^n \times \mathbb{R}$, let $\tilde{\pi}X$ denote the horizontal lift of a vector field X on B^n. Also we denote by $\underline{K}$ the sectional curvature of B^n. Then by direct computation (see Ogiue [1965] or use the Riemannian submersion technique of O'Neill [1966]), we have $K(\tilde{\pi}X, \tilde{\pi}Y) = \underline{K}(X, Y) - 3\eta\left(\nabla_{\tilde{\pi}X}\tilde{\pi}Y\right)^2$ where $\{X, Y\}$ is an orthonormal pair on B^n. Now $g\left(\nabla_{\tilde{\pi}X}\tilde{\pi}Y, \xi\right) = -g\left(\tilde{\pi}Y, \nabla_{\tilde{\pi}X}\xi\right) = g(\tilde{\pi}Y, \phi\tilde{\pi}X) = g(\tilde{\pi}Y, \tilde{\pi}JX)$. Therefore $B^n \times \mathbb{R}$ has constant ϕ-sectional curvature $c = k - 3$

The following is a Sasakian analogue of Schur's Theorem.

Theorem 1.6.19 (cf. [1]). *If the ϕ-sectional curvature at each point of a Sasakian manifold M of dimension ≥ 5 does not depend on the choice of ϕ-section at that point, then it is constant on M.*

The following is the unique existence theorem of Sasakian space forms.

Theorem 1.6.20. *For any two simply connected complete Sasakian manifolds of constant ϕ-sectional curvature c, there exists an isomorphism between them which preserves their almost contact metric structures.*

Chapter 2 CR Manifolds

Note that the wedge $\wedge$ is taken by Alt convention, not Det convention as usual.

§ 2.1 CR manifolds

Let M be a m-dimensional smooth manifold. Let $n \in \mathbb{N}$ be an integer s.t. $1 \leq n \leq [m/2]$.

Definition 2.1.1. A complex subbundle $T_{1,0}(M)$ of the complexified tangent bundle $T(M)\otimes\mathbb{C}$, of complex rank n, is called **an almost complex CR structure** if

$$T_{1,0}(M) \cap T_{0,1}(M) = (0).$$

Here $T_{0,1}(M) = \overline{T_{1,0}(M)}$. The integers n and $k = m - 2n$ are respectively the CR dimension and CR codimension of the almost CR structure and (n, k) is its type.

Remark 2.1.1 (cf. [6]). An almost CR manifold of type $(n, 0)$ is an almost complex manifold.

Definition 2.1.2. An almost CR structure is **formally integrable** if

$$\left[\Gamma^\infty\left(U, T_{1,0}(M)\right), \Gamma^\infty\left(U, T_{1,0}(M)\right)\right] \subseteq \Gamma^\infty\left(U, T_{1,0}(M)\right).$$

If this is the case, $T_{1,0}(M)$ is referred to as a **CR structure** and $(M, T_{1,0}(M))$ is a **CR manifold.**

Definition 2.1.3. A smooth map f between two CR manifolds is called a **CR map** if

$$\left(d_x f\right) T_{1,0}(M)_x \subseteq T_{1,0}(N)_{f(x)},$$

where df is the $\mathbb{C}$-linear extension of the differential map.

Remark 2.1.2. CR manifolds are the objects of a category whose arrows are smooth maps preserving CR structures. The complex manifolds and holomorphic maps form a subcategory of the category of CR manifolds and CR maps.

Let $(M, T_{1,0}(M))$ be a CR manifold of type (n, k).

Definition 2.1.4. The **maximal complex** or **Levi distribution** of $(M, T_{1,0}(M))$ is

$$H(M) = \mathrm{Re}\left\{T_{1,0}(M) \oplus T_{0,1}(M)\right\}.$$

It carries the **complex structure** $J_b : H(M) \to H(M)$ given by

$$J_b(V + \bar{V}) = i(V - \bar{V}).$$

Formally integrable condition is equivalent to

$$\begin{gathered}\left[J_b X, Y\right] + \left[X, J_b Y\right] \in \Gamma^\infty(U, H(M)),\\ \left[J_b X, J_b Y\right] - [X, Y] = J_b\left\{\left[J_b X, Y\right] + \left[X, J_b Y\right]\right\}\end{gathered}$$

CR map is equivalent to say

$$\left(d_x f\right) H(M)_x \subseteq H(N)_{f(x)}$$

and

$$\left(d_x f\right) \circ J_{b,x} = J^N_{b,f(x)} \circ \left(d_x f\right).$$

Definition 2.1.5. $f : M \to N$ is a **CR isomorphism** (or a CR equivalence) if f is a smooth diffeomorphism and a CR map.

CR manifolds arise mainly as real submanifolds of complex manifolds. Let V be a complex manifold, of complex dimension N , and let $M \subset V$ be a real m-dimensional submanifold. Let us set

$$T_{1,0}(M) := T^{1,0}(V) \cap T(M) \otimes \mathbb{C}.$$

Proposition 2.1.6. *If M is a real hypersurface $(m = 2N - 1)$, then$T_{1,0}(M)$ is a CR structure of type $(N - 1, 1)$.*

In general the complex dimension of $T_{1,0}(M)$ defined above may not be constant. Nevertheless, if $\dim_{\mathbb{C}} T_{1,0}(M) \equiv n$, then $(M, T_{1,0}(M))$ is a CR manifold of type (n, k).

Definition 2.1.7. If the CR codimension of $(M, T_{1,0}(M))$ and the codimension of M as a real submanifold of V coincide, i.e., $k = 2N - m$, then $(M, T_{1,0}(M))$ is termed **generic**.

§ 2.1.1 Hypersurface type

Let

$$E_x = \{\omega \in T^*_x(M) : H(M)_x \subset \ker(\omega)\}$$

Assume M is orientable, then the bundle E has a globally defined nowhere vanishing sections $\theta \in \Gamma(E)$. Given any **pseudo-Hermitian structure** θ on M the **Levi form** L_θ is defined by

$$\boxed{L_\theta(Z, \bar{W}) := -i(d\theta)(Z, \bar{W}) = G_\theta(Z, \overline{W}).}$$

Remark 2.1.3. The definition of Levi form of type (n, k) could be possible, cf. [6].

Define a bilinear form (also called **Levi form**)

$$\boxed{G_\theta(X,Y) = (d\theta)\left(X, J_b Y\right),}$$

then

$$\begin{aligned} G_\theta\left(J_b X, J_b Y\right) - G_\theta(X,Y) &= -(d\theta)\left(J_b X, Y\right) - (d\theta)\left(X, J_b Y\right) \\ &= \frac{1}{2}\left\{\theta\left(\left[J_b X, Y\right]\right) + \theta\left(\left[X, J_b Y\right]\right)\right\} \\ &= \frac{1}{2}\theta\left(J_b\left\{[X,Y] - \left[J_b X, J_b Y\right]\right\}\right) = 0. \end{aligned}$$

Define the 2-form Ω by

$$\boxed{\Omega(X,Y) := g_\theta(X, JY) = -d\theta(X,Y).}$$

Definition 2.1.8. We say that $\left(M, T_{1,0}(M)\right)$ is **nondegenerate** if the Levi form L_θ is, the pair (M,θ) is referred to as a **pseudo-Hermitian manifold**, and **strictly pseudoconvex** if L_θ is positive.

Remark 2.1.4. Any two pseudo-Hermitian structure $\theta, \hat{\theta}$ are related by

$$\hat{\theta} = \lambda\theta \tag{2.1}$$

for some non-zero function λ. Then Levi form changes according to

$$L_{\hat{\theta}} = \lambda L_\theta.$$

Hence non-degeneracy is a **CR-invariant** property, i.e., it is invariant under a transformation (2.1).

Definition 2.1.9. We say that a CR map $f : M \to N$ is a **pseudo-Hermitian map** if for some $c \in \mathbb{R}$,

$$f^*\theta = c\theta.$$

If $c = 1$, then f is referred to as an **isopseudo-Hermitian map**.

Definition 2.1.10 (Webster metric).

$$g_\theta(X,Y) = G_\theta(X,Y), \quad g_\theta(X,T) = 0, \quad g_\theta(T,T) = 1$$

$$\boxed{g_\theta = \pi_H G_\theta + \theta \otimes \theta.}$$

Proposition 2.1.11 (Reeb vector field). *There is a unique globally defined nowhere zero tangent vector field T on M such that*

$$\theta(T) = 1, \quad T \rfloor d\theta = 0.$$

CR geometry and contact Riemannian geometry

Let M be a $(2n+1)$-dimensional smooth manifold. Let (ϕ, ξ, η) be a synthetic object, consisting of a $(1,1)$-tensor field $\phi : T(M) \to T(M)$, a tangent vector field $\xi \in \mathcal{X}(M)$, and a differential 1-form η on M. Then (ϕ, ξ, η) is an **almost contact structure** if

$$\phi^2 = -I + \eta \otimes \xi, \quad \phi\xi = 0, \quad \eta(\xi) = 1, \quad \eta \circ \phi = 0.$$

An almost contact structure (ϕ, ξ, η) is said to be **normal** if (recall (1.31))

$$[\phi, \phi] + 2(d\eta) \otimes \xi = 0.$$

A Riemannian metric g on M is said to be **compatible** with the almost contact structure (ϕ, ξ, η) if

$$g(\phi X, \phi Y) = g(X, Y) - \eta(X)\eta(Y).$$

A synthetic object (ϕ, ξ, η, g) consisting of an almost contact structure (ϕ, ξ, η) and a compatible Riemannian metric g is said to be an **almost contact metric structure**.

Given an almost contact metric structure (ϕ, ξ, η, g) one defines a 2-form Ω by setting

$$\Omega(X, Y) = g(X, \phi Y).$$

(ϕ, ξ, η, g) is said to satisfy the **contact condition** if $\Omega = d\eta$, and if this is the case, (ϕ, ξ, η, g) is called a **contact metric structure** on M.

A contact metric structure (ϕ, ξ, η, g) that is also normal is called a **Sasakian structure** (and g is a **Sasakian metric**).

Example 2.1.12. Strictly pseudoconvex CR manifolds are contact Riemannian manifolds, in a natural way. In fact,

$$\begin{aligned} J_b^2 &= -I + \theta \otimes T, \\ J_b T &= 0, \quad \theta \circ J_b = 0, \quad g_\theta(X, T) = \theta(X), \\ g_\theta\left(J_b X, J_b Y\right) &= g_\theta(X, Y) - \theta(X)\theta(Y), \end{aligned}$$

and

$$\Omega = -d\theta.$$

Theorem 2.1.13 ([6]). *Let $(M, T_{1,0}(M), \theta)$ be a pseudo-Hermitian manifold. Then the Webster metric g_θ is Sasakian if and only if the Tanaka-Webster connection ∇ has vanishing* ***pseudo-Hermitian torsion****, that is, $\tau = 0$.*

Remark 2.1.5. The pseudo-Hermitian torsion of ∇, τ is defined by

$$\boxed{\tau(X) = T_\nabla(T, X).}$$

T_∇ is torsion tensor of the Tanaka–Webster connection.

Theorem 2.1.14 (Tanaka-Webster connection on pseudo-Hermitian manifold). *There exists a connection ∇ with the following properties:*
1. $H(M)$ is parallel w.r.t. ∇;
2. $\nabla g_\theta = 0, \nabla J = 0, \nabla\theta = 0, \nabla T = 0$.
3. $T_\nabla(X,Y) = -2\Omega(X,Y)T, T_\nabla(T,JX) = -JT_\nabla(T,X)$.

Remark 2.1.6. We say T_∇ is **pure** if

$$T_\nabla(Z,W) = 0 \tag{2.2}$$
$$T_\nabla(Z,\bar{W}) = 2iL_\theta(Z,\bar{W})T \tag{2.3}$$
$$\tau\circ J + J\circ\tau = 0 \tag{2.4}$$

Theorem 2.1.15 (Tanaka-Webster connection). *Let $(M, T_{1,0}(M), \theta)$ be a nondegenerate CR manifold. Then there is a unique linear connection ∇ on M satisfying the following axioms:*

(i) $H(M)$ is parallel w.r.t. ∇.

(ii) $\nabla J = 0, \quad \nabla g_\theta = 0$.

(iii) T_∇ is pure.

Proof. Let $\pi_\pm : T_{\mathbb{C}}M \to T_{1,0}(M)$ $(T_{0,1}(M))$ be the natural projections. Then $\pi_- Z = \overline{\pi_+ \bar{Z}}$. Note also that

$$T_{\mathbb{C}}M = T_{1,0}(M) \oplus T_{1,0}(M) \oplus \mathbb{C}T.$$

Uniqueness: From (2.3),

$$[\bar{Y}, Z] = \nabla_{\bar{Y}} Z - \nabla_Z \bar{Y} + 2iL_\theta(Z,\bar{Y})T, \quad \forall Y, Z \in T_{1,0}(M).$$

□

Proposition 2.1.16. *The following formulas hold:*

$$\boxed{T_\nabla = 2(\theta\wedge\tau - \Omega\otimes T)}$$

$$\boxed{\nabla^\theta = \nabla + (\Omega - A)T + \tau\otimes\theta + 2\theta\odot J}$$

$$\mathrm{tr}_{g_\theta}(\nabla^\theta - \nabla) = \mathrm{tr}_{G_\theta}(\nabla^\theta - \nabla) = 0.$$

Theorem 2.1.17. *The curvatures of TW connection and LC connection are related by*

$$\begin{aligned} R^\theta(X,Y)Z = R(X,Y)Z &+ (LX\wedge LY)Z - 2\Omega(X,Y)JZ \\ &- g_\theta(S(X,Y),Z)T + \theta(Z)S(X,Y) \\ &- 2g_\theta((\theta\wedge\mathcal{O})(X,Y),Z)T + 2\theta(Z)(\theta\wedge\mathcal{O})(X,Y), \end{aligned}$$

for any $X, Y, Z \in TM$. where

$$S(X,Y) = (\nabla_X\tau)Y - (\nabla_Y\tau)X,$$

$$\mathcal{O} = \tau^2 + 2J\tau - I, \quad L = \tau + J.$$

Remark 2.1.7. Note that

$$J^2Y = -Y + \theta(Y)T, \quad \forall Y \in TM.$$

Theorem 2.1.18 (Structure equations for TW connection).

$$\begin{aligned} d\theta &= 2\sqrt{-1}h_{\alpha\bar\beta}\theta^\alpha \wedge \theta^{\bar\beta}, \\ d\theta^\alpha &= \theta^\beta \wedge \omega^\alpha_\beta + \theta \wedge \tau^\alpha, \quad dh_{\alpha\bar\beta} = \omega^\gamma_\alpha h_{\gamma\bar\beta} + h_{\alpha\bar\gamma}\omega^{\bar\gamma}_{\bar\beta}, \\ d\omega^\alpha_\beta &= -\omega^\alpha_\gamma \wedge \omega^\gamma_\beta + \Pi^\alpha_\beta, \end{aligned} \tag{2.5}$$

where

$$\begin{aligned} \Pi^\alpha_\beta = R^\alpha_{\beta\gamma\bar\delta}\theta^\gamma \wedge \theta^{\bar\delta} + W^\alpha_{\beta\gamma}\theta^\gamma \wedge \theta - W^\alpha_{\beta\bar\gamma}\theta^{\bar\gamma} \wedge \theta \\ + 2\sqrt{-1}\theta_\beta \wedge \tau^\alpha - 2\sqrt{-1}\tau_\beta \wedge \theta^\alpha, \end{aligned} \tag{2.6}$$

and

$$W^\beta_{\alpha\bar\gamma} = h^{\bar\delta\beta}A_{\bar\gamma\bar\delta,\alpha}, \quad W^\beta_{\alpha\gamma} = h^{\bar\delta\beta}A_{\alpha\gamma,\bar\delta}, \quad \tau_\alpha = h_{\alpha\bar\beta}\tau^{\bar\beta}, \quad \theta_\alpha = h_{\alpha\bar\beta}\theta^{\bar\beta}. \tag{2.7}$$

Remark 2.1.8.

$$d\theta^{\bar\alpha} = \theta^{\bar\beta} \wedge \omega^{\bar\alpha}_{\bar\beta} + \theta \wedge \tau^{\bar\alpha} \tag{2.8}$$

§ 2.2 Divergence formulas

Definition 2.2.1. Define the divergence of a vector $X \in TM$ by

$$\operatorname{div}(X)\Psi = \mathcal{L}_X\Psi,$$

where Ψ is a volume form.

Remark 2.2.1. We can take

$$\Psi = \theta \wedge (d\theta)^n.$$

Remark 2.2.2. By 3rd relation of Proposition 2.1.16,

$$\operatorname{div}(X) = \operatorname{tr}_{g_\theta}\{Y \in TM \to \nabla_Y X\}.$$

Note that

$$g_\theta(\nabla^\theta_{e_A}X, e_A) = g_\theta(\nabla^\theta_{T_\alpha}X, T_{\bar\alpha}) + g_\theta(\nabla^\theta_{T_{\bar\alpha}}X, T_\alpha) + g_\theta(\nabla^\theta_T X, T),$$

assuming $\{T_\alpha\}$ is orthonormal and $T_\alpha = \frac{1}{\sqrt{2}}(e_\alpha - iJe_\alpha)$.

Exercise 2.2.2. For any complex vector $Z \in T_{\mathbb{C}}(M)$,

$$\overline{\operatorname{div}(Z)} = \operatorname{div}(\bar Z).$$

Definition 2.2.3. Define the divergence of a 1-form σ by

$$\operatorname{div}(\sigma) = \operatorname{div}(X_\sigma),$$

where X_σ is the dual vector of σ w.r.t. g_θ, i.e., $g_\theta(X_\sigma, Y) = \sigma(Y), \forall Y \in TM$.

Exercise 2.2.4.

$$\operatorname{div}(Z) = T_\alpha(Z^\alpha) + Z^\beta \omega_\beta^\alpha(T_\alpha), \quad \forall Z = Z^\alpha T_\alpha \in T_{1,0}(M).$$

$$\operatorname{div}(\sigma) = \sigma_{\alpha,\bar\beta} h^{\alpha\bar\beta}.$$

Definition 2.2.5. Sublaplacian for functions is defined by

$$\Delta_b(u) := -\operatorname{div}(\nabla^{\mathcal{H}} u) = -(u_{\alpha\bar\beta} h^{\alpha\bar\beta} + u_{\bar\alpha\beta} h^{\bar\alpha\beta}),$$

where $\nabla^{\mathcal{H}} u$ is the horizontal component of the gradient ∇u.

Exercise 2.2.6. The usual Laplacian is given by

$$\Delta u = -(u_{\alpha\bar\beta} h^{\alpha\bar\beta} + u_{\bar\alpha\beta} h^{\bar\alpha\beta} + u_{00}).$$

Chapter 3 Geometry of CR-Submanifolds

§ 3.1 CR-submanifolds and CR-structures

Let N be a n-dimensional *almost Hermitian manifold* with almost complex structure J and with Hermitian metric g. Let M be a *real* m-dimensional Riemannian manifold isometrically immersed in N.

Definition 3.1.1. The differential geometry of M depends on the behaviour of the tangent bundle of M relative to the action of the almost complex structure J. Thus M is called a ***complex*** **(holomorphic)** submanifold if T_xM is invariant by J, i.e., we have

$$J(T_xM) = T_xM, \quad \text{for each } x \in M.$$

Also, we say that M is a ***totally real*** **(anti-invariant)** submanifold of N if we have

$$J(T_xM) \subseteq T_xM^{\perp}, \quad \text{for each } x \in M.$$

These two classes of submanifolds have been extensively investigated in the last decade from different viewpoints.

Definition 3.1.2. In 1978 "we" (refer to the author of [2]) initiated a study of the differential geometry of a new class of submanifolds situated between the above two classes, called CR-submanifolds. More precisely, M is said to be a **CR-submanifold** of N if there exists a differentiable distribution

$$D : x \mapsto D_x \subseteq T_xM,$$

on M satisfying the following conditions:

(i) D is holomorphic, i.e., $J(D_x) = D_x$ for each $x \in M$,

(ii) the complementary orthogonal distribution

$$D^{\perp} : x \mapsto D_x^{\perp} \subseteq T_xM^{\perp},$$

is anti-invariant, i.e., $J(D_x^{\perp}) \subseteq (T_xM)^{\perp}$ for each $x \in M$.

We denote by p the complex dimension of the distribution. D and by q the real dimension of the distribution $D^{\perp}$. Then for $q = 0$ (resp. $p = 0$) a CR-submanifold becomes a **complex submanifold (resp. totally real submanifold)**. If $q = \dim T_x M^{\perp}$ the CR-submanifold is called an **anti-holomorphic submanifold**. A **proper CR-submanifold** is a CR-submanifold which is neither a complex submanifold nor a totally real submanifold.

Example 3.1.3. Each real hypersurface M of N (with $n \geq 2$) is a proper CR-submanifold. In fact we define

$$D_x^{\perp} : x \mapsto D_x^{\perp} = J(T_x M^{\perp})$$

and take D as the complementary orthogonal distribution to $D^{\perp}$ in TM. Thus M is endowed with a pair of distributions $(D, D^{\perp})$ satisfying the conditions of the definition of a CR-submanifold. Moreover, we have $\dim_{\mathbb{R}} D_x^{\perp} = 1$ and $\dim_{\mathbb{C}} D_x = n - 1$.

Hence, M is a proper CR-submanifold.

Remark 3.1.1. It is easily seen that on a CR-submanifold the distribution D (resp. $D^{\perp}$) is the maximal distribution invariant by J (resp. anti-invariant by J), i.e., if D' (resp. D'') is an invariant (resp. anti-invariant) distribution on M, then we have $D'_x \subseteq D_x$ (resp. $D''_x \subseteq D_x$), for each $x \in M$.

Definition 3.1.4. Now let M be an arbitrary Riemannian manifold isometrically immersed in an almost Hermitian manifold N. For each vector field X tangent to M we put

$$JX = \phi X + \omega X, \tag{3.1}$$

where ϕX and ωX are respectively the tangent part and the normal part of JX. Also, for each vector field V normal to M we put

$$JV = BV + CV, \tag{3.2}$$

where BV and CV are respectively the tangent part and the normal part of JV.

Theorem 3.1.5. *The submanifold M of N is a CR-submanifold if and only if we have*

$$rank(\phi) = constant, \omega\phi = 0.$$

or, if and only if

$$rank(B) = constant, \phi B = 0.$$

Remark 3.1.2. Note that

$$\begin{aligned} \phi X &= JPX \\ \omega X &= JQX \end{aligned} \tag{3.3}$$

for any $X \in TM$, P, Q is the projection to $D, D^{\perp}$ respectively.

Also,

$$\begin{aligned} D &= Im(\phi) = ker(\omega) \\ D^{\perp} &= Im(B) = ker(\phi). \end{aligned} \tag{3.4}$$

Remark 3.1.3. Now, suppose M is a CR-submanifold of the almost Hermitian manifold N.

$$\begin{aligned} \phi^2 &= -P \\ \phi^3 + \phi &= 0 \\ C^3 + C &= 0. \end{aligned} \tag{3.5}$$

Proposition 3.1.6. *On each CR-submanifold M the vector bundle morphisms ϕ and C define f-structures on TM and $T^{\perp}M$ respectively.*

Theorem 3.1.7 (Blair-Chen). *A CR-submanifold of a **Hermitian** manifold is a CR-manifold.*

§ 3.2 f-structure

Let M_n be an n-dimensional differentiable manifold of class C^∞ and let there be given a tensor field $f \neq 0$ of type $(1,1)$ and of class C^∞ such that

$$f^3 + f = 0. \tag{3.6}$$

Theorem 3.2.1 ([7]). *For a tensor field $f \neq 0$ satisfying (3.6), the operators*

$$l = -f^2, \quad m = f^2 + 1, \tag{3.7}$$

1 *denoting the identity operator, applied to the tangent space at a point of the manifold are complementary projection operators.*

Moreover,

$$fl = lf = f, \tag{3.8}$$

$$fm = mf = 0, lm = ml = 0, \tag{3.9}$$

$$f^2 l = -l, \tag{3.10}$$

$$f^3 m = 0, \tag{3.11}$$

that is, f acts on L as an almost complex structure operator and on M as null operator.

Remark 3.2.1. Thus if there is given a tensor field $f \neq 0$ satisfying (3.6), then there exist complementary distributions L and M corresponding to the projection operators l and m respectively.

Moreover, one can show that if the rank of f is constant and is equal to r, then the dimensions of L and M are r and $n-r$ respectively. We call such a structure an f-structure of rank r.

Let me mention that

Theorem 3.2.2 ([7]). *A necessary and sufficient condition for an n-dimensional manifold to admit a tensor field $f \neq 0$ of type $(1,1)$ and of rank r such that $f^3 + f = 0$ is that r is even: $r = 2m$ and the group of tangent bundle of the manifold be reduced to the group $U(m) \times O(n-2m)$.*

§ 3.3 Integrability of Distributions on a CR-Submanifold

First of all, we have

$$[\phi, \phi](X, Y) = [\phi X, \phi Y] + \phi^2[X, Y] - \phi([\phi X, Y]) - \phi([X, \phi Y]). \tag{3.12}$$

and

$$[J, J](X, Y) = [\phi, \phi](X, Y) - \Omega([\phi X, Y] + [X, \phi Y]). \tag{3.13}$$

Theorem 3.3.1 (Bejancu). *Let M be a CR-submanifold of an almost Hermitian manifold N. Then the distribution D is integrable if and only if*

$$[J, J](X, Y)^{\perp} = 0 \tag{3.14}$$

and

$$Q(\phi, \phi)(X, Y) = 0, \textit{ for any } X, Y \in \Gamma(D). \tag{3.15}$$

Corollary 3.3.2. *Let M be a CR-submanifold of a* ***Hermitian*** *manifold N. The distribution D is integrable if and only if the Nijenhuis tensor of ϕ vanishes identically on D.*

Theorem 3.3.3. *Let M be a CR-submanifold of an almost Hermitian manifold N. The distribution D is integrable if and only if the Nijenhuis tensor of ϕ vanishes identically on $D^{\perp}$.*

§ 3.4 CR-submanifold of a nearly Kaehlerian manifold

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§ 3.5 ϕ-Connections on a CR-Submanifold and CR-Products of Almost Hermitian Manifolds

Definition 3.5.1. A linear connection ∇ on M is called a **ϕ-connection** if ϕ is covariantly constant with respect to this connection, that is, we have

$$\nabla_X \phi = 0, \forall X \in TM \tag{3.16}$$

Theorem 3.5.2. *Both distributions D and $D^{\perp}$ on a CR submanifold of an almost Hermitian manifold are parallel with respect to any ϕ-connection.*

Definition 3.5.3. We say that a CR-submanifold M of an almost Hermitian manifold N is a **CR-product** if both distributions D and $D^{\perp}$ are integrable and M is locally a Riemannian product $M_1 \times M_2$, where M_1 is a leaf of D and M_2 is a leaf of $D^{\perp}$. A CR-product with $D \neq \{0\}$ and $D^{\perp} \neq \{0\}$ is called a **proper CR-product.**

Example 3.5.4 (An counterexample). There exist no proper CR submanifolds of S^6 with both distributions D and $D^\perp$ parallel with respect to Levi-Civita connection. However, Sekigawa constructed an example of a 3-dimensional proper CR-submanifold such that both distributions D and $D^\perp$ are integrable.

Theorem 3.5.5. *Let M be a CR-submanifold of an almost Hermitian manifold N. If the Levi-Civita connection on M is a ϕ-connection, then M is a CR-product.*

Recall that

Theorem 3.5.6. *Both distributions D and $D^\perp$ are parallel with respect to Levi-Civita connection if and only if they are integrable and their leaves are totally geodesic in N.*

§ 3.6 CR-submanifolds of Kaehlerian manifolds

Theorem 3.6.1. *Let M be a CR-submanifold of a Kählerian manifold N. Then we have*

(i) the distribution $D^\perp$ is integrable,

(ii) the distribution D is integrable if and only if the second fundamental form of M satisfies

$$h(X, JY) = h(Y, JX), \text{ for any } X, Y \in \Gamma(D).$$

Remark 3.6.1. Blair and Chen have constructed an example of a CR-submanifold of a Hermitian manifold for which $D^\perp$ is not integrable.

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